



Estimating whole-of-event consequences in interdependent infrastructure systems

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ABSTRACT

Conventional risk assessments underestimate the total burden of hazards by characterizing consequences at or near the point of disruption, rather than as processes that unfold over time. This paper addresses that limitation by introducing a method to estimate whole-of-event consequences in interdependent infrastructure systems, integrating cascading failures and recovery dynamics into time-dependent risk metrics. We develop a modeling framework that generates trajectories of service disruption and restoration across electricity, water, and wastewater networks, and quantify community-level consequences using metrics such as property-days without service, outage duration thresholds, and spatial disparities in recovery. Applied to a coastal flooding case study, the results show that recovery-informed estimates differ substantially from exposure-based and static cascading approaches, revealing both greater total impacts and different spatial distributions of disruption. These findings demonstrate that a substantial portion of risk arises from the accumulation of disruption during recovery, which is not captured in conventional assessments. Whole-of-event consequences therefore provide a more appropriate basis for understanding systemic risk and for informing infrastructure investment, adaptation planning, and risk prioritization. By operationalizing consequence as a time-dependent, system-wide process, this work advances risk assessment practice and supports more accurate and equitable decision-making in interdependent infrastructure systems.

1. Introduction

Conventional risk assessments systematically underestimate the total burden of hazards because they characterize consequences at or near the point of disruption, rather than as processes that unfold over time within interdependent systems [1–3]. As a result, risk is often misrepresented, overlooking how consequences accumulate, propagate, and persist across systems and communities. While many models estimate the likelihood and immediate impacts of hazards, they typically truncate analysis at the moment of failure, omitting the temporal and systemic dynamics that shape real-world outcomes. In complex infrastructure networks, the duration and distribution of service loss are influenced not only by initial damage but also by the structure of interdependencies and the process of recovery [4,5]. These time-dependent consequences, including how long communities go without essential services and where cascading failures prolong disruption, are central to understanding risk in modern systems. Yet, few existing approaches provide a way to estimate whole-of-event consequences that

capture the combined effects of time, interdependence, and recovery across infrastructure and communities [6,7].

In modern infrastructure systems, disruptions rarely remain isolated because services such as electricity, water, and wastewater are deeply interdependent [4,5,8]. Failures in one network often propagate into others; for example, the loss of electricity during a storm may render water pumping stations or wastewater treatment plants inoperable. These cascading effects extend disruption beyond the initially affected assets, shaping both the scale and spatial distribution of service loss. Capturing these higher-order failures is therefore essential for understanding how disruption unfolds across interconnected systems, particularly when assessing risk to communities [9,10]. Yet many existing models either isolate infrastructure systems or fail to represent how interdependence shapes the progression of disruption over time [7,11,12].

The consequences of disruption are not determined at the point of failure, but by how disruption evolves and is resolved over time.

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Beyond the initial loss of services, the recovery period plays a crucial role in shaping the consequences that communities experience. The duration of service loss can influence whether residents can continue to live and function in their communities, maintain employment, or access essential services, or whether prolonged outages result in displacement [13]. Risk is therefore inherently time-dependent, and incorporating outage duration into assessments is essential to capture the full scale of impacts experienced over the event [2]. This temporal dimension reveals how disruption accumulates and may disproportionately affect certain groups, supporting more equitable risk treatment in infrastructure planning and adaptation decisions [14,15]. In interdependent infrastructure systems, these dynamics are further amplified. When systems such as electricity, water, and wastewater are interlinked, the restoration of one network often depends on the functionality of another [10]. As a result, recovery is not independent across systems but unfolds through interconnected pathways that can delay or redistribute restoration. Yet, in practice, recovery modeling frequently overlooks these dependencies [7]. The literature remains dominated by single-infrastructure studies focused on estimating expected outage durations within individual networks, including electricity [16–22], transportation [23–25], and water [26–32]. These approaches do not account for how disruption cascades across services or how recovery dynamics shape consequences over time.

While recovery modeling has advanced in the context of response and restoration optimization, it has not been integrated into risk assessment in a way that captures the full consequences of disruption. Current approaches typically characterize consequence at or near the point of impact, treating it as a static outcome rather than a process that unfolds over time [1,2]. In doing so, they overlook how disruption accumulates, propagates across interdependent systems, and is shaped by recovery dynamics, which are often amplified in interconnected infrastructure networks [4,10,12,33]. Although recovery is often analyzed in terms of system performance, such as time-to-repair or network functionality, these representations do not capture the total burden experienced by communities, including prolonged service loss, displacement, and uneven spatial impacts [5,7,8]. As a result, existing methods, whether focused on cascading failures, resilience, or recovery optimization (e.g., [34,35]), do not estimate the cumulative, community-level consequences that unfold over the full course of an event. This gap reflects the absence of approaches that integrate time, interdependence, and recovery into consequence estimation, despite their central role in shaping risk in complex systems [2,3,9].

To address this gap, this paper introduces a method to estimate whole-of-event consequences by integrating interdependence and recovery into time-dependent risk metrics, enabling direct comparison with conventional exposure-based and static cascading assessments. We then investigate the following research questions:

- **RQ1:** How can whole-of-event consequences be estimated by representing restoration dynamics in interdependent infrastructure systems?
- **RQ2:** How does the timing and duration of restoration influence the severity, equity, and spatial distribution of community-level consequences?
- **RQ3:** How do whole-of-event consequence estimates differ from those based on exposure or immediate outages alone?

We respond to these questions by applying a whole-of-event consequence estimation framework to a real-world urban case study, evaluating disruption from a coastal flooding event. The framework simulates restoration across electricity, water, and wastewater systems, accounting for direct damage, interdependencies, and recovery trajectories.

While our implementation uses mixed-integer programming to represent constraints and sequencing, we intentionally abstract from operational decision-making to generate internally consistent recovery trajectories for consequence estimation. These trajectories enable the

estimation of whole-of-event consequences, including metrics such as property-days without service, long-duration outages, and spatial disparities in restoration. In doing so, this paper demonstrates how consequence estimation, and therefore risk assessment for planning, changes when disruption is treated as a dynamic, system-wide process rather than a static outcome.

2. Methods

Our objective is to estimate whole-of-event consequences in interdependent infrastructure systems by capturing how disruption unfolds and is resolved over time. To do this, we generate time-dependent trajectories of system disruption and recovery, which enable consequences to be represented as cumulative, system-wide processes rather than static outcomes.

We develop an integrated modeling framework that combines cascading failure simulation with restoration modeling to represent how disruptions propagate across interdependent networks and how recovery dynamics shape system behavior over time. These trajectories enable the estimation of consequences as experienced by communities, including the duration of service loss, prolonged outages, and spatial disparities in recovery.

The workflow links hazard exposure and infrastructure data to a sequence of system states over time, from initial disruption through to recovery. These system trajectories form the basis for estimating consequence metrics that reflect whole-of-event consequences. The following subsections describe the modeling workflow (Section 2.1), current approaches to modeling recovery (Section 2.2), the representation of interdependent infrastructure systems (Section 2.3), the restoration model (Section 2.4), the metrics used to quantify consequences relevant to risk assessment (Section 2.5), and the application to a real-world case study (Section 2.6).

2.1. Overview and modeling workflow

To estimate whole-of-event consequences across interdependent systems, we develop a modeling workflow that generates time-dependent trajectories of disruption and recovery. This workflow integrates a cascading failure model with a restoration model to represent how disruptions propagate across infrastructure networks and are resolved over time.

The workflow is shown in Fig. 1. Infrastructure assets and hazard layers are used to simulate immediate and cascading failures, which define the initial system state. These are then used to generate recovery trajectories that capture how services are restored under interdependencies and resource constraints. The resulting sequence of system states over time forms the basis for estimating whole-of-event consequences.

2.2. Recovery modeling in interdependent systems

To estimate whole-of-event consequences, we draw on existing approaches for modeling recovery in interdependent systems. These approaches are typically designed to simulate restoration processes or optimize recovery decisions; here, we use them to generate time-dependent system trajectories for consequence estimation.

Prior studies have used a range of methods to simulate disruption and restoration across multiple infrastructure systems, including network-based models [6,36–41], inoperability input–output models [36,42–46], Bayesian networks [47–50], agent-based models [51, 52], and machine learning models [53,54]. Game-theoretic approaches, including Stackelberg formulations, have also been used to model cascading failures and recovery dynamics in interdependent infrastructure systems (e.g., [55–57]), capturing strategic interactions between system components and decision-makers. These methods differ in how

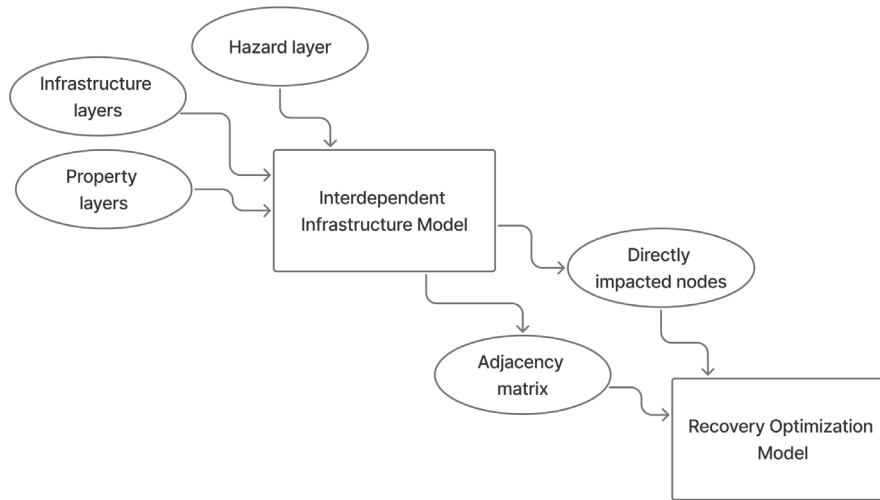


Fig. 1. Workflow showing how infrastructure and hazard data are used to generate time-dependent system trajectories, which are then used to estimate consequences.

they represent system interdependencies, uncertainty, and recovery processes.

To simulate restoration, studies usually employ optimization-based methods (e.g., mixed-integer programming), decision-making rules (e.g., in agent-based models), probabilistic sampling (e.g., sampling from conditional probability tables in a Bayesian network or from a restoration or recovery curve), or data-driven methods (e.g., machine learning) [58,59]. Among these, mixed-integer programming (MIP) is frequently used to determine restoration sequences [37,60–62], minimize the cost of damage and losses [63], or stay within certain budget constraints [64]. Other applications of MIP have incorporated parameters and constraints that reflect the decision-making process [65,66], or provide insight to decision-makers by comparing different strategies to restore infrastructure, such as hardening infrastructure or repairing infrastructure in a particular order [36,67].

While most applications focus on optimizing operational recovery, we instead use these methods as a diagnostic tool to trace how disruption unfolds over time and space. This enables the estimation of human-centered consequence metrics such as property-days without service or long-duration outages. Accordingly, our focus is not on identifying optimal recovery strategies, but on quantifying how recovery dynamics shape the magnitude and distribution of consequences.

2.3. Interdependent infrastructure model

To generate the initial disruption state for consequence estimation, we use the network topology-based interdependent infrastructure model from Brunner et al. [6]. The model represents dependencies between infrastructure systems and properties using geospatial infrastructure and hazard data.

Disruption is simulated by overlaying hazard layers (e.g., flood depths), applying vulnerability functions to determine asset functionality, and propagating failures through the network based on interdependencies. This produces both directly impacted nodes and cascading failures across interconnected systems. This initial disruption state forms the starting point for modeling recovery trajectories and estimating whole-of-event consequences.

The infrastructure systems considered include the electrical power network, water supply network, and wastewater network; each property node requires all three to be considered fully functional. For simplicity, direct exposure is used to determine whether an impacted node is functional or not.

2.4. Restoration model

To generate recovery trajectories for consequence estimation, we simulate infrastructure restoration over discrete time periods using a mixed-integer linear program implemented in Pyomo [68,69] and solved with Gurobi [70]. This builds on the interdependent infrastructure model of Brunner et al. [6] (which represents direct and cascading failures), by introducing the restoration model to generate time-dependent system trajectories for consequence estimation. The model tracks the functional state of each node (utility or property) across time, subject to resource constraints and interdependencies between infrastructure systems. Restoration requires both the repair of damaged nodes and the availability of upstream services, such that service can only be restored when all required dependencies are functional.

This formulation is used to generate time-dependent system trajectories that capture how disruption is resolved across interdependent networks. These trajectories form the basis for estimating whole-of-event consequences.

The full model formulation, including sets, parameters, decision variables, and constraints, is provided in [Appendix](#).

2.5. Consequence metrics

To estimate whole-of-event consequences, we define a set of metrics that quantify the magnitude, duration, and spatial distribution of disruption experienced by communities. These metrics are evaluated under three representations of consequence: (i) hazard exposure, (ii) initial service outage including cascading effects, and (iii), whole-of-event consequences that incorporate recovery dynamics over time.

Consequence perspective 1: Hazard exposure

This represents a conventional static assessment based solely on physical exposure to the hazard.

- **Number and percentage of properties exposed to the hazard** based on spatial overlap with flood depth thresholds.

Consequence perspective 2: Initial service outage (static cascading)

This view includes cascading infrastructure failure at the time of impact (t_0), but excludes restoration dynamics.

- **Number and percentage of properties without power, water, or wastewater at t_0 .**

- **Number and percentage of properties without any service at t_0 .**
- **Number and percentage of properties without all three services (multi-service deprivation) at t_0 .**

Consequence perspective 3: Whole-of-event consequences

This perspective accounts for infrastructure restoration dynamics, interdependencies, and outage duration, and forms the basis for estimating cascading risk.

- **Property-days without service**, calculated for each utility and aggregated across time.
- **Number and percentage of properties exceeding outage duration thresholds** (e.g., longer than seven days).
- **Time to 90% recovery**, aggregated by spatial unit.
- **Maximum outage duration** per utility and spatial unit.
- **Multi-utility deprivation**: properties without all three services, or any service, for extended durations.

Equity and spatial metrics

To assess the distribution of consequences (RQ2), we compute:

- **Ranked score comparison plots** to compare property-level burden between static cascading and whole-of-event perspectives.
- **Ridgeline plots** of outage durations for each utility to characterize temporal distributions.
- **Choropleth maps** of key metrics to identify spatial patterns in service loss and recovery.

Together, these metrics enable comparison between exposure-based and time-integrated representations of disruption, highlighting how consequence estimates change when recovery dynamics and interdependencies are explicitly incorporated [71–74].

2.6. Case study

To demonstrate the estimation of whole-of-event consequences in an applied setting, we implement the modeling framework using a real-world case study of coastal flooding. The approach is generalizable to any hazard and infrastructure context where geospatial exposure, interdependencies, and recovery processes can be represented.

The case study is based on Ōtautahi Christchurch in Aotearoa New Zealand, a low-lying coastal city with interconnected river and estuarine systems. Coastal flooding is represented using hazard layers that estimate water depths for a specified annual exceedance probability (AEP) and sea level rise (SLR) scenario. To account for uncertainty in the hazard model, flood depths of 10 cm or less are excluded. This threshold is consistent with standard practice in New Zealand flood modeling, where shallow depths are often associated with rain-on-grid artifacts and transient surface water rather than sustained ponding that would affect infrastructure functionality. However, the sensitivity of results to alternative depth thresholds is an important consideration and is identified as an area for future work.

The analysis uses one property dataset and three infrastructure networks (electricity, water supply, and wastewater) with dependencies both within and across systems. The utility assets were provided by the respective infrastructure operators, while the property dataset is based on water meter locations across the study area (see [6]). Properties are assumed to depend on all three utilities to be fully functional. Where asset dependencies are not explicitly defined, connections are inferred using proximity along the road network. As a byproduct of this approach, some utility nodes do not have any property dependencies for this case study.

This case builds on Brunner et al. [6], enabling comparison between models that consider only initial cascading failures and those that incorporate recovery dynamics.

3. Results

Our analysis examines to what extent risk is misestimated when consequences are truncated at the point of disruption, and how incorporating restoration dynamics changes the estimation of whole-of-event consequences in interdependent infrastructure systems (RQ1–RQ3). The results are organized to first characterize system-level restoration patterns, then to compare consequence estimates across different representations of disruption (RQ3), and finally to examine the timing, duration, and spatial distribution of impacts (RQ2). Together, these analyses demonstrate how recovery-informed modeling alters both the magnitude and distribution of estimated consequences.

3.1. Utility network restoration

We begin by examining the restoration of the electricity, water, and wastewater networks following a 1% AEP coastal flooding event under current sea level. These network-level results provide context for understanding how restoration dynamics translate into property-level consequences.

We parameterize the recovery model based on the following assumptions:

- Each timestep represents one day.
- Ten repair crews (which can work on any utility) are available each day.
- Each damaged asset requires two repair crews for a single day to be repaired.
- A minimum of one and a maximum of two repair crews are needed to start work on an asset.

These parameters are specified to generate internally consistent recovery trajectories rather than to represent site-specific operational conditions. The aim is to isolate how restoration dynamics influence consequence estimation, rather than to optimize or predict real-world recovery processes.

The restoration of each utility network is shown in Fig. 2 as a function of total system performance. These values represent assets without service and include both those directly exposed to floodwaters and those made non-functional due to upstream failures. In this event, all damaged infrastructure nodes were restored within 45 days. The wastewater network (green line) experienced the greatest proportional loss of service, followed by the water network (blue line) and the electricity network (red line). No utility experienced more than a 20% loss in service. The water and wastewater networks recovered sharply within the first ten days, while the electricity network shows a fairly constant restoration trend. These differences partially reflect network size: the electricity network comprises many more assets than the water supply or wastewater networks. These system-level restoration patterns define the temporal structure through which disruption propagates and is resolved, forming the basis for estimating whole-of-event consequences.

3.2. Comparing consequence estimates across risk estimation approaches

To address RQ3, we compare how risk is represented under three approaches that differ in how they account for disruption over time: (1) hazard exposure only, (2) static cascading outages at the time of impact (t_0), and (3) recovery-informed, whole-of-event consequences. This comparison allows us to quantify how much of the total burden is missed when recovery dynamics are excluded.

Fig. 3 provides an overview of each of these metrics. The yellow bar shows the number of properties that were directly exposed to the coastal flooding event. The colored dots at t_0 represent initial outages, accounting for the network dependencies, and show the number of properties without any of the three utilities (black), without electricity

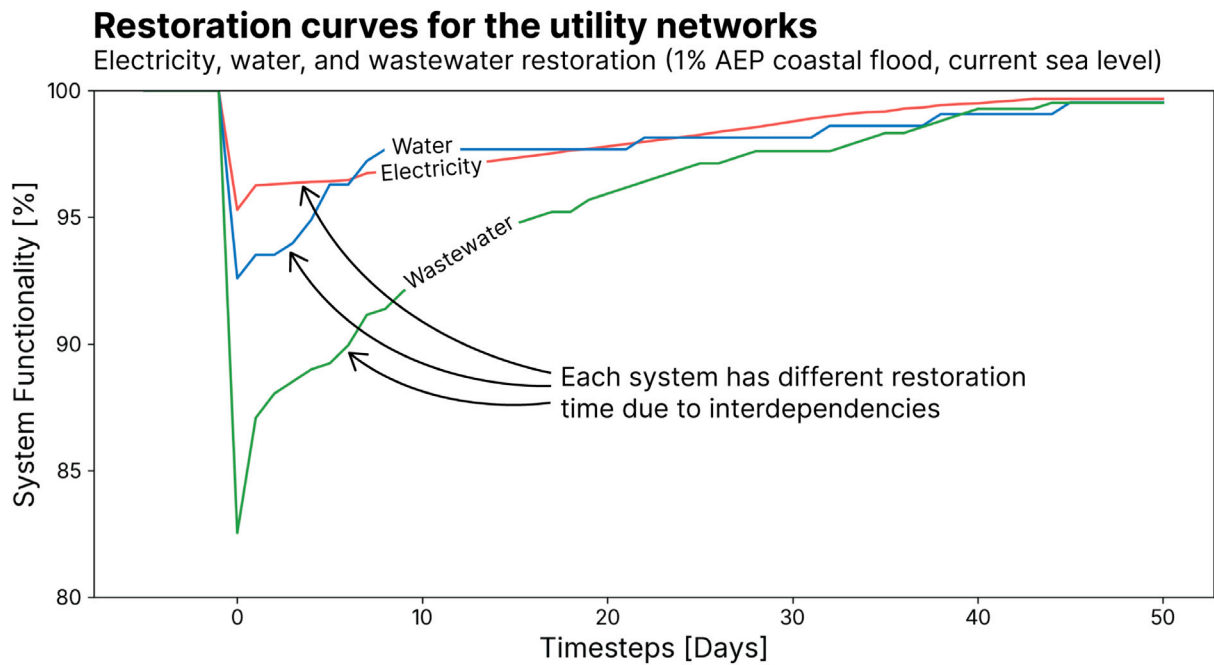


Fig. 2. Restoration of each utility network over time as a function of total system performance for electricity (red), water (blue), and wastewater (green). These trajectories define how disruption evolves and is resolved across interdependent systems, forming the temporal basis for estimating whole-of-event consequences.

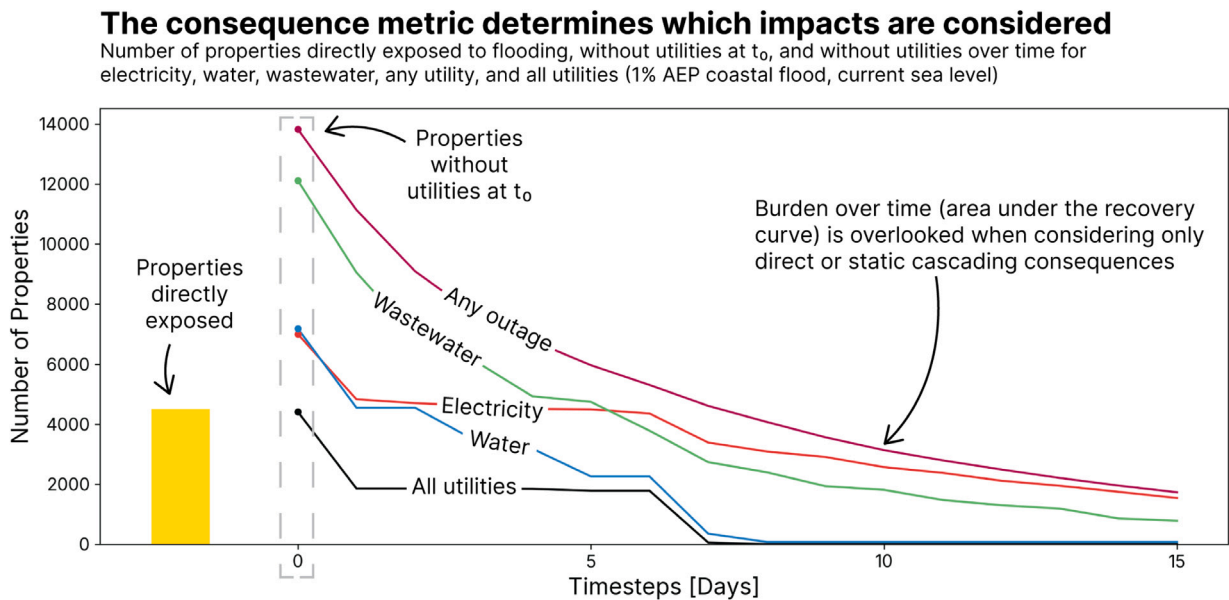


Fig. 3. Comparison of consequence representations: directly exposed properties (yellow bar), initial cascading outages at t_0 (colored dots), and recovery-informed trajectories over time (colored lines). The divergence between the point-in-time estimates and the recovery trajectories illustrates how static approaches underestimate both the duration and total magnitude of service disruption.

(red), without water (blue), without wastewater (green), and with any outage (maroon). The lines show recovery-informed trajectories, with all properties fully restored after 38 days (in contrast to full utility restoration after 45 days). Note that this difference in restoration time is due to the automated approach of linking the properties to their closest utility node, which, for this application, leaves a small number of utility nodes without property dependencies. The relative slopes and durations reflect the network restoration patterns in Fig. 2, but translated to property-level consequences.

Fig. 4 maps these first two perspectives: directly exposed properties and utilities (left), and initial utility loss at t_0 (right). These figures highlight how cascading effects extend the footprint of disruption beyond

the initial hazard inundation zone. This expansion of the disruption footprint indicates that even at the point of impact, exposure-based assessments fail to capture the full extent of service loss. The corresponding metrics are listed in Table 1, including property-days without service for the whole-of-event perspective. These values quantify the cumulative burden over time and show that exposure-based and static cascading approaches systematically underestimate total consequences. Recovery-informed modeling captures tens of thousands of additional property-days of outage that are not visible in point-in-time assessments, demonstrating that a substantial portion of risk arises from the accumulation of disruption during recovery.

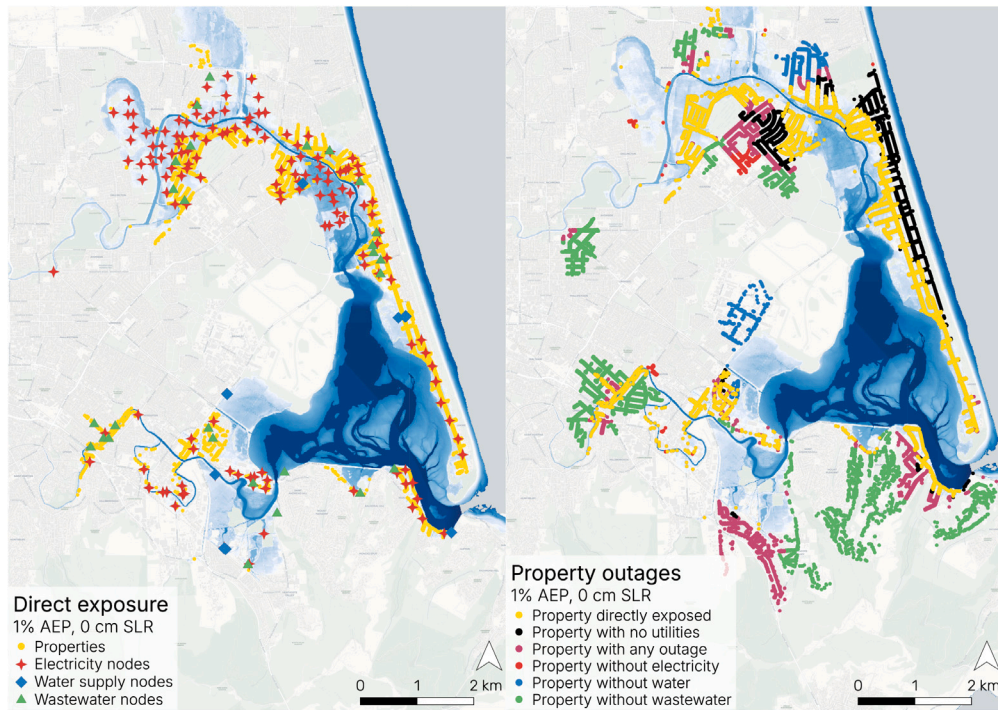


Fig. 4. Maps showing the directly exposed properties and utilities (left) and the initial utility loss per property at t_0 (right). Cascading failures extend disruption beyond the hazard footprint, demonstrating that exposure-based assessments underestimate the spatial extent of service loss.

We also examine how property prioritization changes between static and recovery-informed approaches. For each property, we compare rankings derived from static cascading conditions and from whole-of-event consequences. The static ranking reflects initial exposure and outages at t_0 , while the recovery-informed ranking reflects cumulative outage burden over time. Properties are ranked under each representation and compared to identify how prioritization changes when recovery dynamics are included. This gives a value that is:

- Equal to 1 when the property has the same static cascading rank as the recovery rank,
- Greater than 1 when the static cascading rank is higher than the recovery rank, and
- Less than 1 when the static cascading rank is lower than the recovery rank.

Ranking properties by each score and plotting them against each other based on their rank values (Fig. 5, left) reveals significant differences in prioritization. Properties near the diagonal (having a ratio between 0.99 and 1.01) have similar ranks in both views, while those far above or below it are over- or under-prioritized when using static metrics alone. Fig. 5 (right) shows where these changes occur spatially, based on the ratio between static cascading and recovery score ranks.

Overall, this comparison shows that excluding recovery dynamics understates both the magnitude and the distribution of disruption, and can lead to different prioritization of affected properties and communities.

3.3. Temporal and spatial dynamics of recovery

To address RQ2, we examine how the timing and duration of restoration shape the magnitude and spatial distribution of whole-of-event consequences. The maps show outputs based on the H3 hexagon grid, which provides a consistent equal-area spatial representation, in contrast to census-based units that vary in size to reflect population distribution. Here, we focus on three key outage metrics mapped at the hexagon level: (1) the maximum outage duration among properties in

Table 1

Comparison of consequence metrics across exposure-based, static cascading, and recovery-informed approaches for a 1% AEP coastal flooding event. Recovery-informed metrics reveal substantially higher cumulative disruption, demonstrating that point-in-time approaches systematically underestimate whole-of-event consequences. Proportions at t_7 are calculated relative to the number of properties in each respective outage category at t_0 .

Directly exposed properties	4509
Properties without electricity at t_0	7000
Properties without water at t_0	7185
Properties without wastewater at t_0	12,120
Properties with any outages at t_0	13,827
Proportion of properties with any outage at t_7	38.4%
Proportion of properties without electricity at t_7	62.2%
Proportion of properties without water at t_7	31.4%
Proportion of properties without wastewater at t_7	31.2%
Property-days without electricity	63,378
Property-days without water	28,889
Property-days without wastewater	65,478
Property-days with any outage	94,474
Property-days without all three utilities	15,448

the hexagon, (2) the time until 90% of properties in the hexagon have all utilities restored, and (3) the number of properties in the hexagon still without all utilities after seven days.

These maps (Fig. 6) show that the most severely impacted areas include the spit between the estuary and eastern coast, and properties along the two rivers. There are also isolated patches of prolonged outages driven by the exposure of a small number of critical assets, though in most cases these are fixed within the first seven days. The first metric highlights locations where at least one property experiences a long recovery tail; the second shows areas where the majority of properties take longer to be restored; and the third captures short-term persistence of service loss in the initial recovery phase. Here it is presented as the number of affected properties, which reflects the scale of disruption and supports prioritization decisions. A proportional

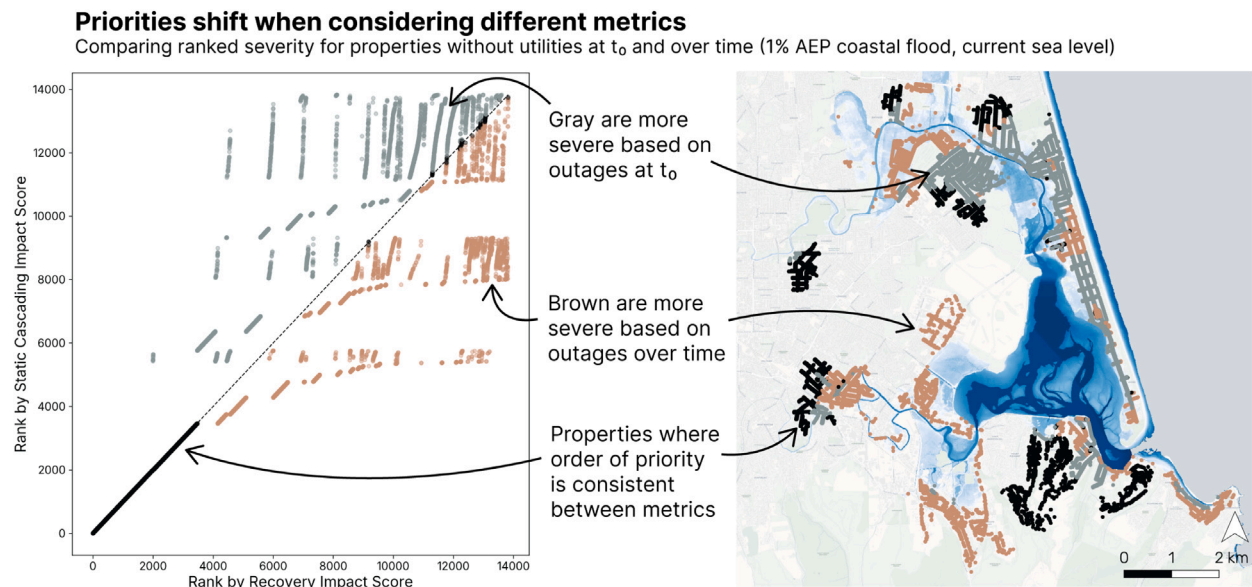


Fig. 5. Comparison of property-level rankings based on static cascading outages and recovery-informed consequence metrics (left), and spatial distribution of ranking differences (right). Differences in ranking demonstrate how excluding recovery dynamics can misidentify which properties and communities experience the greatest cumulative burden.

Spatial distribution of burden based on different recovery metrics

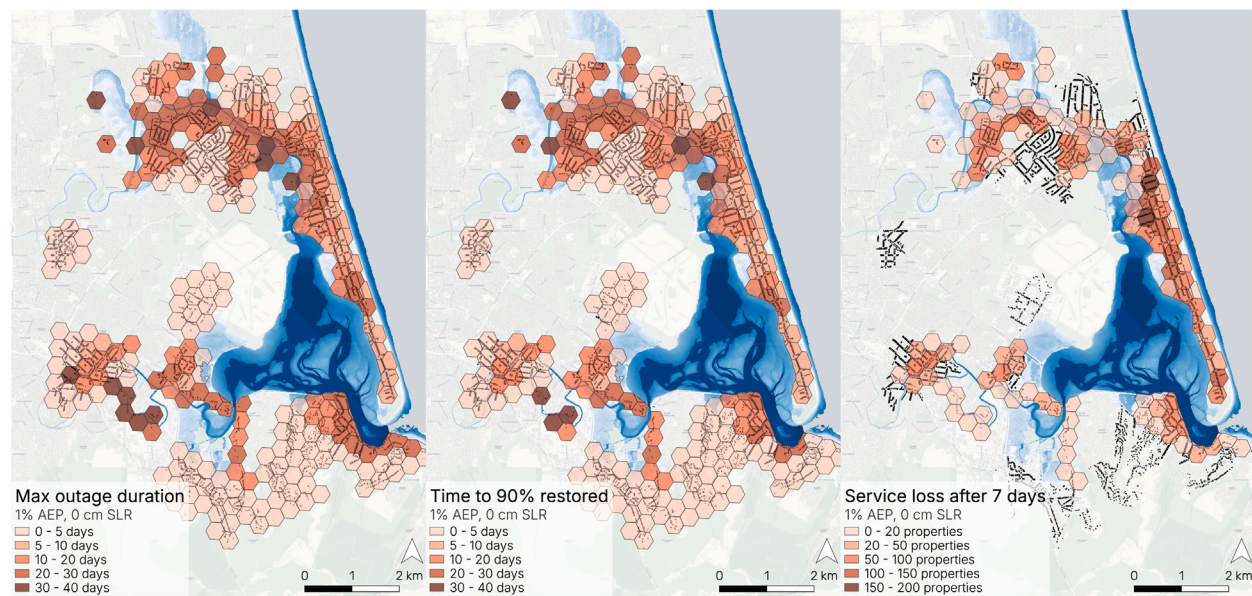


Fig. 6. Spatial distribution of recovery-informed consequence metrics: maximum outage duration (left), time to 90% recovery (center), and properties without full service after seven days (right). These maps show how restoration dynamics redistribute impacts across space and time, revealing patterns of prolonged disruption not captured in static assessments.

representation would provide a complementary, density-independent view of impact and can be readily derived within the same framework.

The distribution of outage durations for each utility is summarized in Fig. 7 as a ridgeline plot. This graphic shows that many properties have short outages of just one day, reflecting quick, early gains in network functionality. Water supply outages (blue) mostly resolve within seven days, while wastewater (green) exhibits a steadier recovery after a slight peak at seven days. Electricity outages (red) similarly show an initial peak at seven days, but a larger proportion of properties remain affected for a longer duration compared to the other utilities.

These temporal and spatial patterns demonstrate that the severity of disruption is not determined solely by initial damage, but by

how disruption evolves over time. Areas with extended restoration timelines experience disproportionately higher cumulative burdens, and differences in recovery rates across utilities redistribute impacts across communities. These dynamics are not captured in static assessments, but are central to estimating whole-of-event consequences in risk analysis.

4. Discussion

Our results show that risk is systematically underestimated when consequences are truncated at the point of impact. By incorporating recovery dynamics and interdependencies, whole-of-event consequence

Infrastructure interdependence leads to variable outage durations

The distribution of properties' restoration time, by utility (1% AEP coastal flood, current sea level). This shows how infrastructure dependence influences restoration patterns

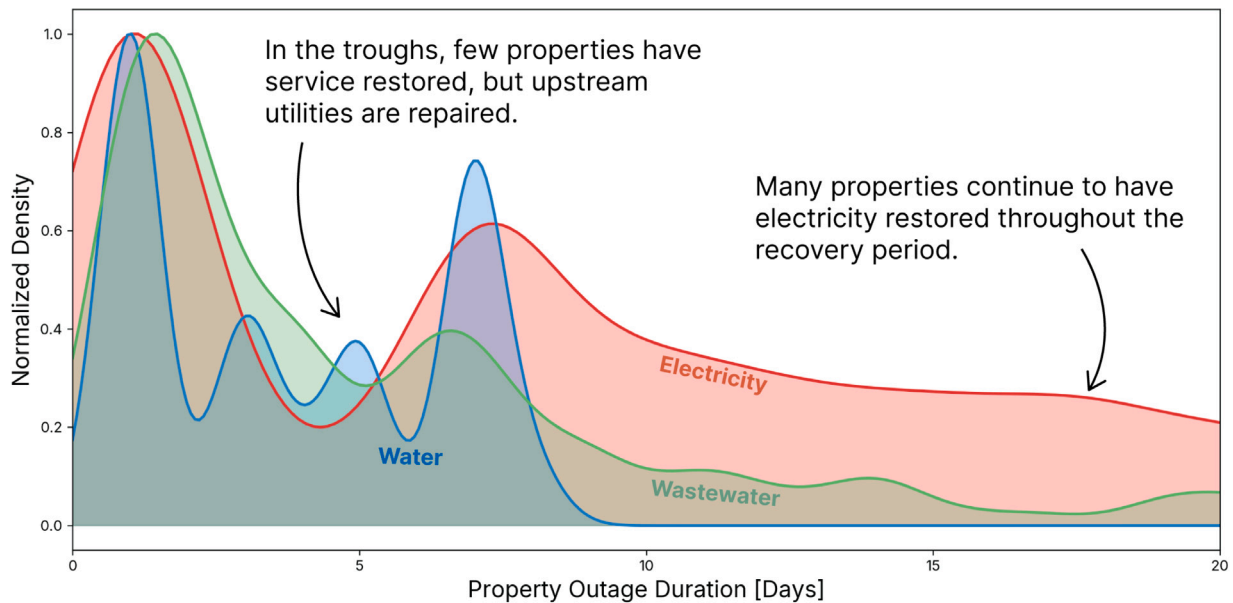


Fig. 7. Distribution of outage durations for electricity (red; $n = 7000$), water (blue; $n = 7185$), and wastewater (green; $n = 12,120$). n is the number of properties initially without service. Differences in duration across utilities highlight how recovery dynamics shape the temporal accumulation of disruption, contributing to whole-of-event consequences.

estimation reveals substantially greater and differently distributed impacts across space and time (RQ3). Our results show that both the magnitude and spatial distribution of disruption change when consequences are treated as cumulative processes rather than static outcomes. Recovery trajectories further demonstrate that outage duration varies widely across services and locations, shaping inequities in how disruption is experienced (RQ2). In the remainder of this section, we interpret these findings in the context of RQ1, considering how recovery-informed modeling improves consequence estimation in risk assessment.

4.1. Consequence estimates differ when time is included

Our results show that exposure-based and static cascading approaches systematically under-represent the full extent of disruption. Comparing perspectives that include recovery modeling reveals not only higher total impacts but also different spatial and temporal patterns of harm.

For example, in Table 1 the static cascading perspective (properties experiencing outages at t_0) shows similar numbers for water and electricity at around 7000 properties each. When recovery duration is included, these consequences are no longer similar: water totals around 29,000 property-days without service, whereas electricity exceeds 63,000 property-days without service. This shift in relative burden has implications for restoration priorities, suggesting they might be strategically re-ordered to reduce prolonged outages.

The map of the static cascading-to-recovery score rank ratio (Fig. 5) highlights where these differences matter geographically. Properties shaded gray are over-prioritized when using static cascading scores, while the brown properties are under-prioritized relative to their recovery burden. If the recovery score is treated as the more comprehensive metric, the static approach risks diverting attention to the gray properties at the expense of brown ones that experience longer outages. Clusters of gray and brown in Fig. 5 show that these misalignments are spatially patterned, not random.

The choice of consequence metric therefore shapes both the magnitude and geography of perceived risk, which has implications for

emergency management in the immediate recovery period (e.g., where to send resources, who may need temporary housing) and for long-term planning (e.g., which assets or areas to strengthen, where to guide development). These spatial patterns of over- and under-prioritization underscore the role of outage duration and recovery timing in shaping the severity and equity of impacts across the city.

These differences highlight a fundamental distinction between response-oriented and planning-oriented metrics. Response metrics emphasize the speed of restoration, whereas planning requires understanding the total burden experienced over the event, including duration and inequity of disruption. Whole-of-event consequences therefore provide a more appropriate basis for infrastructure investment, adaptation planning, and risk prioritization.

4.2. Recovery dynamics shape the severity and distribution of impacts

Including recovery trajectories reveals how whole-of-event consequences are unevenly distributed across space, time, and infrastructure services. These inequities are illustrated in Fig. 6, which maps three whole-of-event metrics by grouping properties into hexagons: the maximum outage durations within the hexagon, the time until 90% of properties regain all utilities, and the number of properties still without all utilities after seven days. Areas along the northern river corridor experience the longest outages across all utilities, with both the maximum duration and the time to 90% recovery extending well beyond other parts of the city. The southern river corridor has similarly long outage durations, but far fewer properties still without all utilities after seven days. In contrast, the eastern coastal area shows a much higher proportion of properties with ongoing outages at day seven. More affluent neighborhoods in the southern part of the map experience outages that are generally shorter and less extensive than those in less affluent areas such as the northern river corridor and the eastern coast.

Such recovery-based outcomes provide a more tangible measure of human-centric impacts, making them better suited to inform decision-makers and communities [75]. They can also guide resource allocation by identifying where outages persist beyond defined thresholds and

how many people are affected. While much of the focus in recovery planning is on the infrastructure itself [76], understanding the implications of the restoration process through duration-based metrics provides better insight into the community impacts. The inclusion of duration-based metrics aligns with concepts such as “planning emergency levels of service”, where utility operators provide guidelines for expected outage durations during emergencies [77]. Previous work has applied recovery dynamics to analyze risk to infrastructure [78]; here, we present a methodology that extends the use of recovery dynamics to community risk assessments. These recovery dynamics present a more socially relevant perspective of risk, which can help guide equitable planning in adaptation [79]. This lays the groundwork for further investigation into detailed socio-economic and demographic analysis to understand which groups are particularly burdened, be it through extended outages or number of impacted utilities, and how recovery prioritization may be adjusted to address these disparities. These metrics directly inform planning decisions by identifying where prolonged disruption occurs and which communities experience the greatest cumulative burden.

4.3. Advancing risk science through time-dependent consequence estimation

Risk assessments have traditionally focused on exposure at the point of impact, implicitly assuming that consequences are static. However, consequences evolve over time through cascading failures and recovery processes, meaning that risk is fundamentally a function of how disruption accumulates and is resolved [1–3,9]. This study addresses this limitation by operationalizing consequence as a time-dependent process and demonstrating how whole-of-event consequences can be estimated in practice.

Through the coastal flooding case study, we showed how the number of properties impacted, the duration of outages, and the spatial extent of service loss change depending on the snapshot in time chosen. When recovery is omitted, or only represented from an infrastructure-centric perspective, the resulting estimates understate the burden of disruption and can misdirect both immediate emergency response and long-term adaptation planning. Recovery-based metrics reveal that resilience can be strengthened (and risk reduced) not only before an event through asset hardening, but also during and after the event by reducing outage durations and targeting restoration to the most affected communities.

While the magnitude of impacts depends on model assumptions, the key findings are structural. Integrating consequences over time necessarily increases the total estimated burden, and interdependencies redistribute that burden across space and services. These effects arise from the structure of the system rather than specific parameter choices, and are therefore expected to be directionally robust across different contexts and modeling approaches.

These findings extend prior work on interdependent infrastructure risk modeling [78,80] by shifting the focus from asset performance to community-centered consequences, and by quantifying how those consequences change over time. This work responds directly to calls in the risk science literature for integrating time into risk assessment [2] and for moving beyond static hazard-exposure-vulnerability framings to account for the dynamics of systemic risk [3,4,81,82]. By showing how recovery modeling can be used diagnostically to produce these time-dependent, socially relevant metrics, this study contributes both a methodological framework and a conceptual shift for risk assessment.

While the case study focuses on three utility networks, the framework can incorporate additional infrastructure systems such as telecommunications, stormwater, and transport. Including these networks would capture further layers of interdependence and enable the analysis of indirect effects such as functional isolation (where people can physically reach amenities that are themselves non-functional [80]) and the ability of amenities to reopen following natural hazards [83]. The

approach can also incorporate partial functionality, varying restoration times by damage extent, and refined vulnerability functions to improve realism. In addition, integrating social vulnerability and resilience metrics would provide a more comprehensive understanding of how different population groups are affected by prolonged service disruption and their capacity to respond and recover, building on the consequence metrics developed in this study.

While the current application focuses on a coastal flooding scenario, the restoration framework is structure-agnostic but requires hazard-specific parameterization to define the initial disruption state. Application to other hazards (e.g., earthquakes, storm surge, or multi-hazard events) would involve adapting the hazard intensity inputs, vulnerability functions, and resulting initial failure states, while the restoration and consequence estimation framework remains unchanged. The hazard extents of, for example, utility outages from seismic events or from a series of hazard events (for multi-hazard analysis) could be mapped out and input into the initial infrastructure model which feeds through the hazard agnostic restoration framework.

The mixed-integer programming formulation used here is one of several approaches that can generate recovery trajectories under constraints. While alternative methods (e.g., inoperability input–output models, Bayesian networks, or agent-based models) may offer different representations of recovery processes, the consequence metrics developed in this study are not dependent on the specific modeling technique and can be applied across methods. This implementation has several limitations, including computational requirements for large networks, limited adaptability during simulation, and sensitivity to input assumptions. Future work could explore alternative modeling approaches and uncertainty analysis to further assess robustness, while retaining the consequence estimation framework presented here.

In sum, this work demonstrates that risk is systematically misestimated when consequence is treated as a static outcome. By integrating recovery dynamics and interdependencies, the proposed framework provides a practical approach to estimating whole-of-event consequences, revealing both greater total impacts and different spatial distributions of risk. These metrics offer a more appropriate basis for equitable and resilient infrastructure planning, supporting decisions on investment, adaptation, and risk prioritization.

5. Conclusion

Accurate risk assessment requires representing consequences as processes that unfold over time, rather than as static outcomes at the point of impact. This study addresses that requirement by developing and applying a whole-of-event consequence estimation framework that integrates hazard exposure, interdependent network failure, and restoration dynamics to produce time-dependent, community-centered metrics.

Applied to a coastal flooding case study, the framework demonstrates that the scale, severity, and distribution of impacts depend not only on where failures occur, but on how disruption unfolds and is resolved. Metrics such as property-days without service, outage duration thresholds, and multi-service deprivation provide a more complete representation of systemic risk than exposure-only or static outage estimates, and reveal inequities in how disruption is experienced.

These whole-of-event consequences provide a more appropriate basis for planning decisions, including infrastructure investment, adaptation prioritization, and risk reduction, compared to response-oriented metrics that emphasize restoration speed. By integrating time into consequence estimation, this work responds to calls in the risk science literature [1–3] to move beyond static hazard-exposure-vulnerability framings.

More broadly, this study shows that risk is not defined at the moment of disruption, but by the cumulative consequences experienced over the course of an event. Capturing these whole-of-event consequences is therefore essential for accurately assessing risk in interdependent systems and for supporting more effective and equitable planning decisions.

CRediT authorship contribution statement

Logan G. Brunner: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daphne Skipper:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Tom M. Logan:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Data curation, Conceptualization.

Disclaimer

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Appendix. Restoration model

This appendix provides the formal specification of the restoration model used to generate recovery trajectories for consequence estimation.

We begin with model notation. Table A.2 defines the model sets, which encode much of the problem structure, including the nodes, with their interdependencies and damages, as well as the number of time periods to include. Time periods may represent any length of time (e.g., hours, days, months, etc.), but must be consistent throughout the model. Table A.3 defines the parameters, which are user-defined (or sometimes industry-defined) numerical inputs to the model, such as the number of repair crews available and how many repair crew-time periods are required to repair an asset. Tables A.4 and A.5 define the decision variables, which track the recovery process over time, such as which nodes are functional, which nodes are still damaged, and the number of repair crews assigned by the model to repair each node during each time period.

The model is as follows:

$$\text{maximize} \quad \left(1000 \sum_{i \in T} \sum_{j \in N^{\text{property}}} x_{i,t} \right) + \left(10 \sum_{i \in T} \sum_{j \in D} x_{j,t} \right) + \sum_{i \in T} \sum_{k \in N} x_{k,t}$$

Table A.2

Sets.

N	:=	The set of all network nodes (utilities and properties)
U	:=	The set of node types ($U := \{\text{property, electricity, water, wastewater}\}$)
N^u	:=	The set of nodes of type $u \in U$ ($N^u \subseteq N$)
C^u	:=	The set of “children” served by any node of type $u \in U$ ($C^u \subseteq N$)
P_i^u	:=	The set of type $u \in U$ “parent” utility nodes that serve node $i \in N$ ($P_i^u \subseteq N^u$)
D	:=	The set of damaged nodes in need of repair ($D \subseteq N$)
D^u	:=	The set of damaged nodes of type $u \in U$ in need of repair ($D^u \subseteq D$)
T	:=	The set of discrete, fixed-length time periods

$$+ \left(-0.1 \sum_{i \in T} \sum_{j \in D} w_{j,t} \right), \quad (\text{A.1})$$

$$\text{subject to} \quad \sum_{i \in D^u} w_{i,t} \leq b_{u,t}, \quad \forall t \in T, u \in U; \quad (\text{A.2})$$

$$\ell_i z_{i,t} \leq w_{i,t} \leq u_i z_{i,t}, \quad \forall i \in D, t \in T; \quad (\text{A.3})$$

$$z_{i,t-1} - z_{i,t} \leq y_{i,t-1} + q_{i,t}, \quad \forall i \in D, t \in T \setminus \{1\}; \quad (\text{A.4})$$

$$c_i y_{i,t} \leq w_{i,1} + \sum_{t'=2}^t (w_{i,t'} - p_i q_{i,t'}), \quad \forall i \in D, t \in T; \quad (\text{A.5})$$

$$\sum_{i \in T} y_{i,t} \leq 1, \quad \forall i \in D; \quad (\text{A.6})$$

$$x_{i,t} \leq \sum_{t'=1}^t y_{i,t'}, \quad \forall i \in D, t \in T; \quad (\text{A.7})$$

$$x_{i,t} \leq \sum_{j \in P_i^u} x_{j,t}, \quad \forall (u,i) : u \in U, i \in C^u; t \in T; \quad (\text{A.8})$$

$$x_{i,t} \in \{0, 1\}, \quad \forall i \in N, t \in T; \quad (\text{A.9})$$

$$y_{i,t}, z_{i,t} \in \{0, 1\}, \quad \forall i \in D, t \in T; \quad (\text{A.10})$$

$$q_{i,t} \in \{0, 1\}, \quad \forall i \in D, t \in T \setminus \{1\}; \quad (\text{A.11})$$

$$w_{i,t} \in \mathbb{Z}^{\geq 0}, \quad \forall i \in D, t \in T. \quad (\text{A.12})$$

The objective function (A.1) has four main terms, the first of which represents the primary objective of restoring as many properties to a fully functioning state (having service from all utility types) as early in the recovery period as possible; the other three terms support the expected behavior of the restoration process:

- First term: (coefficient of 1000) ensures that the primary focus of optimization is to return as many properties to the fully functioning state (having all utilities) as early in the recovery horizon as possible.
- Second term: (coefficient of 10) encourages all damaged utility nodes to be repaired, even if all properties already have functional utilities.
- Third term: (coefficient of 1) similar to the second term, encourages full functionality for undamaged utility nodes.
- Fourth term: (coefficient of -0.1) discourages the assignment of unneeded repair crews.

The coefficients are specified to guide the sequencing of restoration under constraints, rather than to represent calibrated decision preferences. The optimization is computationally intensive so reaching a single, optimal solution for the case study is not feasible. The inclusion of these coefficients helps to reduce the objective space, thus enabling a reasonable level of optimality after several hours to days of simulation time. The coefficient 1000 is a reward for functional properties, so that the main priority is to restore as many properties as possible. The next term has a coefficient of 10, which is a reward for all repaired nodes — after all the utility nodes with downstream properties are restored, it will then repair the other utility nodes that have no connections. The coefficient of 1 is to simply ensure that all nodes (utility and property) that are undamaged and without dependencies will be set by default

Table A.3

Parameters.

c_i	:=	The number of crew-time periods (e.g., crew-days) required to repair damaged node $i \in D$
ℓ_i, u_i	:=	The minimum and maximum, respectively, number of crews that can be assigned to repair node $i \in D$ during a time period when i is under repair ($u_i \geq \ell_i > 0$)
p_i	:=	The number of extra crew-time periods required to repair node $i \in D$ each time work is interrupted (i.e., each time crews are moved off node i for one or more time periods after the repair to i has begun)
$b_{u,t}$	:=	The number of crews available in time period $t \in T$ to work on utility type $u \in U$

Table A.4

Binary (0/1) decision variables.

$x_{i,t}$	:=	1 if node $i \in N$ is fully functional during time $t \in T$; 0 otherwise
$y_{i,t}$	:=	1 if repair on node $i \in D$ is completed during time $t \in T$; 0 otherwise
$z_{i,t}$	:=	1 if at least ℓ_i workers are assigned to node $i \in D$ during time $t \in T$; 0 otherwise
$q_{i,t}$	:=	1 if no workers are assigned to $i \in D$ during time $T \setminus \{1\}$ after workers were assigned to i during time $t-1$ but i was not fully repaired during time $t-1$, 0 otherwise

Table A.5

Integer decision variables.

$w_{i,t}$	:=	The number of workers assigned to $i \in D$ during time $t \in R$
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to “functional” in the restoration simulation. Finally, the -0.1 is to penalize any workers assigned, which is enough to discourage assigning workers after every node has been restored. The main reason for these coefficients is because we are not finding the single optimized solution but are using an MIP gap to reduce computational load and time.

Turning to the constraints, the first four manage the assignment of repair crews to jobs. Constraint (A.2) sets the number of repair crews per utility type for each time period. Constraint (A.3) enforces the minimum and maximum number of repair crews assigned to each job during each time period and links the w and z variables; that is, z is 0 if no workers are assigned and 1 if repair crews are assigned. Constraint (A.4) requires that if repair crews are assigned to a job in one time period but not the next, then either the job is complete or a job-interruption penalty will be applied.

The next two constraints identify when a repair is complete. Constraint (A.5) encourages a non-preemptive recovery process to avoid crews switching nodes before they are fully repaired. It counts up the number of crew-time periods that a job has been worked to determine when a job is finished; a crew-time period deduction of size p_i is applied for each interruption to work on the job. In reality, this could be considered the additional time required if a new crew must take over the halfway-repaired node, which the MIP will avoid if p_i is a value above zero. Constraint (A.6) requires that each job is marked “complete” during a single time period (if at all).

The next two constraints identify when a node is fully functioning. Constraint (A.7) requires that a damaged node must have been repaired to be fully functional. Constraint (A.8) requires that every node must have service from each of its required utilities (i.e., at least one of its suppliers of each utility is functional).

The last four constraints, (A.9)–(A.12), define variable types.

These relations have been coded into the optimization module of the Python-based interdependent infrastructure model, which passes in the network adjacency matrix and the nonfunctional nodes to create the sets listed in Table A.2. Additional computational variables for the solver include the number of CPUs, the RAM limit, and the MIP gap. The gap indicates how close the solution is to the optimal solution. The MIP gap for the case study detailed earlier was kept at the default of 0.01%, which took several hours when using a server with 23 CPUs and 120 GB of memory.

Data availability

The authors do not have permission to share data.

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